

Chapter 4

Advanced Diffusion Concepts

Formula-Formatted Exam Study Guide

Main focus. ELBO and noise-prediction MSE, score fields, reverse diffusion as drift plus stochasticity, classifier guidance, classifier-free guidance, noise-aware classifiers, guidance tradeoffs, and FID.

Formatting update. All major formulas and equations are rewritten in mathematical display form using proper fractions, square roots, subscripts, superscripts, expectations, gradients, norms, Gaussian notation, and trace notation.

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0, t, \epsilon} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$

Use the formula sheet first, then read the concept sections, then practice the sample questions and flashcards.

1. What You Must Be Able to Explain

By the end of this chapter, you should be able to explain the following ideas in a clear exam answer:

- why error minimization in reverse diffusion is connected to the ELBO;
- what the score function means geometrically;
- why noise prediction is closely related to score estimation;
- how reverse diffusion combines score-guided drift with calibrated noise;
- why guided diffusion is needed for conditional generation;
- the difference between classifier guidance and classifier-free guidance (CFG);
- why classifier guidance needs a classifier trained on noisy states and timestep information;
- how guidance scale changes realism, controllability, diversity, and artifacts;
- how FID measures feature-distribution matching and why it is incomplete.

2. One-Page Formula Sheet

Concept	Formatted equation
Forward transition	$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t I), \quad \beta_t = 1 - \alpha_t$
Closed-form noising	$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$
Noise-prediction loss	$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0, t, \epsilon} \left[\ \epsilon - \epsilon_{\theta}(x_t, t)\ _2^2 \right]$
Reverse model	$p_{\theta}(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$
Reverse mean using ϵ_{θ}	$\mu_{\theta}(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon_{\theta}(x_t, t) \right)$
Score definition	$s_t(x) = \nabla_x \log p_t(x)$
Gaussian score	$p(x) = \mathcal{N}(\mu, \sigma^2) \implies s(x) = -\frac{x - \mu}{\sigma^2}$

Standard Gaussian score	$p(x) = \mathcal{N}(0, 1) \implies s(x) = -x$
Denoising-score identity	$s_t(x_t) \approx -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}$
Score-form reverse step	$x_{t-1} = x_t + \beta_t s_t(x_t) + \sqrt{\beta_t} z, \quad z \sim \mathcal{N}(0, I)$
Noise-form reverse step	$x_{t-1} \approx x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) + \sqrt{\beta_t} z$
Bayes basis for guidance	$\nabla_x \log p(x y) = \nabla_x \log p(x) + \nabla_x \log p(y x)$
Classifier guidance	$s_t^{\text{guided}}(x_t) = s_t(x_t) + w \nabla_{x_t} \log p_\phi(y x_t, t)$
Classifier-free guidance	$\epsilon_{\text{CFG}} = (1 + w) \epsilon_\theta(x_t, t, y) - w \epsilon_\theta(x_t, t, \emptyset)$
Fréchet Inception Distance	$\text{FID} = \ \mu_r - \mu_g\ _2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$

3. ELBO, Error Estimation, and Why MSE Appears

Diffusion models can be trained from a probabilistic objective. Ideally, we want to maximize the marginal likelihood of the clean data, but this likelihood is intractable because it requires integrating over the full diffusion trajectory. Therefore, we optimize a tractable variational lower bound:

$$\log p_\theta(x_0) \geq \mathcal{L}_{\text{ELBO}}.$$

The diffusion ELBO contains a reconstruction term, intermediate KL terms, and a prior-matching term. When the forward posterior is Gaussian and the reverse variance is fixed appropriately, the KL terms reduce to matching Gaussian means. Rewriting the reverse mean using the predicted noise gives the familiar noise-prediction MSE:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0, t, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \right].$$

Important exam point. Noise prediction is not just an engineering trick. It is a practical form of variational inference because minimizing the noise-prediction error optimizes the ELBO up to constants and weighting factors.

4. Score Function: The Geometry of Probability

The score is the gradient of the log-density. It tells us the local direction in which probability increases most rapidly:

$$s(x) = \nabla_x \log p(x).$$

For a Gaussian distribution, the score points back toward the mean. For a multimodal distribution, the score points toward the most plausible nearby mode. This is why the score can be interpreted as a denoising direction.

$$p(x) = \mathcal{N}(\mu, \sigma^2) \implies s(x) = -\frac{x - \mu}{\sigma^2}, \quad p(x) = \mathcal{N}(0, 1) \implies s(x) = -x.$$

For a two-mode mixture, each component contributes according to its responsibility:

$$r_k(x) = \frac{p_k(x)}{p_1(x) + p_2(x)}, \quad s(x) = \sum_k r_k(x) \left[-\frac{x - \mu_k}{\sigma^2} \right].$$

5. Noise Prediction Equals Score Estimation

In DDPM training, the neural network is trained to predict the forward noise. Under the Gaussian corruption model, predicting noise is mathematically connected to estimating the score of the noisy

distribution:

$$\begin{aligned}x_t &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, & \epsilon &\sim \mathcal{N}(0, I), \\ \epsilon_\theta(x_t, t) &\approx \mathbb{E}[\epsilon \mid x_t], \\ s_t(x_t) &= \nabla_{x_t} \log p_t(x_t) \approx -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}.\end{aligned}$$

This identity is the key bridge between DDPM noise prediction and score-based diffusion. Even when the model is trained as a noise predictor, it is effectively learning the score field at each noise level.

6. Reverse Diffusion as Drift + Calibrated Noise

Reverse diffusion is not only denoising and not only random sampling. It combines a deterministic drift term and a stochastic noise term:

$$x_{t-1} = x_t + \beta_t s_t(x_t) + \sqrt{\beta_t} z, \quad z \sim \mathcal{N}(0, I).$$

The drift term $\beta_t s_t(x_t)$ moves the sample toward higher-probability regions. The random term $\sqrt{\beta_t} z$ preserves diversity and keeps the chain sampling from a distribution instead of collapsing to a single mode.

$$x_{t-1} \approx x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) + \sqrt{\beta_t} z.$$

Exam wording. The reverse step subtracts the predicted noise direction and adds schedule-controlled Gaussian noise.

7. Guided Diffusion: Why It Is Needed

An unconditional diffusion model learns to generate plausible data. Many applications need controlled samples, such as a chosen class label, text prompt, style, segmentation map, or safety constraint. Guided diffusion modifies the reverse vector field so that samples remain realistic and satisfy a condition:

$$\text{unconditional: } p(x), \quad \text{guided: } p(x \mid y).$$

The central idea is: instead of flowing toward any plausible image, the sample flows toward plausible images consistent with the condition.

8. Classifier Guidance

Classifier guidance uses an unconditional diffusion model plus a separate classifier. By Bayes' rule, the conditional score can be written as an unconditional score plus a classifier gradient:

$$\begin{aligned}\log p(x \mid y) &= \log p(x) + \log p(y \mid x) - \log p(y), \\ \nabla_x \log p(x \mid y) &= \nabla_x \log p(x) + \nabla_x \log p(y \mid x), \\ s_t^{\text{guided}}(x_t) &= s_t(x_t) + w \nabla_{x_t} \log p_\phi(y \mid x_t, t).\end{aligned}$$

Here, $s_t(x_t)$ keeps the sample realistic, while $\nabla_{x_t} \log p_\phi(y \mid x_t, t)$ pushes it toward the desired class or condition. The guidance strength w controls how strongly the condition influences sampling.

Why the classifier must be noise-aware

During reverse diffusion, the classifier is asked to evaluate noisy intermediate states x_t , not only clean images x_0 . A clean-image classifier may give unreliable gradients on noisy inputs. A noise-aware classifier is trained using the same forward diffusion process:

$$p_\phi(y \mid x_t, t).$$

The timestep t must be provided because the meaning of the visible image features depends strongly on the noise level.

9. Classifier-Free Guidance (CFG)

Classifier-free guidance avoids training an external classifier. Instead, one diffusion model learns both conditional and unconditional predictions. During training, the condition is randomly replaced by a null token $\emptyset$ with probability p_{drop} :

$$\epsilon_\theta(x_t, t, y) \quad \text{and} \quad \epsilon_\theta(x_t, t, \emptyset).$$

At sampling time, the two predictions are combined as

$$\epsilon_{\text{CFG}} = (1 + w)\epsilon_\theta(x_t, t, y) - w\epsilon_\theta(x_t, t, \emptyset).$$

This formula amplifies the direction that separates the conditional prediction from the unconditional prediction. That is why CFG usually improves condition adherence, but high w can reduce diversity or create artifacts.

10. Hyperparameters and Tradeoffs

- **Guidance strength w :** larger w improves condition alignment, but may reduce diversity and introduce artifacts.
- **Number of steps T :** fewer steps give faster sampling, but weaker fidelity and less opportunity for guidance to correct the sample.
- **Noise schedule $\beta_t, \alpha_t, \bar{\alpha}_t$:** controls how the signal-to-noise ratio changes across time and affects training stability and sampling quality.
- **Model capacity:** a weak unconditional denoiser cannot be fully repaired by guidance.
- **Classifier robustness:** classifier guidance works only if the classifier gives meaningful gradients on noisy x_t states.
- **Condition-drop probability p_{drop} :** in CFG it determines how often the model learns unconditional behavior.

11. Evaluation: FID

Fréchet Inception Distance compares real and generated samples in a deep feature space. It summarizes both distributions with means and covariances and computes a distance between them:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right).$$

Lower FID usually suggests that generated samples better match the feature statistics of real samples. However, FID is only a proxy: it depends on the feature extractor, sample size, domain match, numerical stability, and it does not fully measure prompt alignment or controllability.

12. Sample Exam Questions with Strong Answers

Question 1: What is the theoretical basis of classifier guidance?

Classifier guidance follows from Bayes' rule. Taking the gradient with respect to x gives the conditional score as the sum of the unconditional realism score and the classifier gradient that pushes the sample toward the desired condition:

$$\nabla_x \log p(x | y) = \nabla_x \log p(x) + \nabla_x \log p(y | x).$$

Question 2: Why is a noise-aware classifier preferable to a standard classifier in classifier-guided diffusion?

The reverse chain queries the classifier on noisy states x_t , not clean images x_0 . A standard classifier trained only on clean data can give unstable or meaningless gradients at high noise levels. A noise-aware classifier learns class evidence across noise levels:

$$p_\phi(y | x_t, t).$$

Question 3: What is the core idea behind classifier-free guidance, and why is random condition dropping essential?

CFG trains one model to produce both conditional and unconditional predictions. Random condition dropping teaches the model a true unconditional branch. At sampling time, the conditional direction is amplified relative to the unconditional reference:

$$\epsilon_{\text{CFG}} = (1 + w)\epsilon_{\text{cond}} - w\epsilon_{\text{uncond}}.$$

Question 4: Why does stronger guidance usually improve condition adherence but reduce diversity?

Larger guidance scale pushes the sampler more strongly toward regions satisfying the condition. This improves alignment but narrows the set of accepted trajectories, so diversity drops and artifacts may appear.

Question 5: What does FID measure conceptually, and what are its limitations?

FID measures how close real and generated feature distributions are by comparing their means and covariances in a deep feature space:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}).$$

It is useful but incomplete because it depends on feature extractor choice, sample size, domain match, and numerical stability.

13. Deduplicated Flashcard Review

This section keeps the flashcard content in question-answer form. Cards containing equations use formatted mathematical notation.

Flashcard 1

Q: Why is the ELBO optimized in diffusion models instead of the marginal likelihood?

A: Because the marginal likelihood is computationally intractable, so the ELBO gives a tractable lower-bound objective.

$$\log p_\theta(x_0) \geq \mathcal{L}_{\text{ELBO}}$$

Flashcard 2

Q: In the diffusion ELBO, what does the prior-matching term at $t = T$ represent?

A: It measures how well the final noised distribution matches the Gaussian prior.

$$q(x_T | x_0) \approx p(x_T) = \mathcal{N}(0, I)$$

Flashcard 3

Q: What simplification allows the diffusion KL terms to reduce to an MSE objective?

A: Fixing the reverse variance and using the Gaussian forward posterior makes each KL term reduce to a mean-matching error.

$$\mathcal{L}_{\text{simple}} = \mathbb{E} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$

Flashcard 4

Q: What is the mathematical definition of a score in diffusion modeling?

A: The score is the gradient of the log probability density.

$$s(x) = \nabla_x \log p(x)$$

Flashcard 5

Q: Geometrically, what does the score function represent?

A: It is the slope of the log-density curve and points toward higher probability.

Flashcard 6

Q: For a standard Gaussian distribution, what is the simplified score function?

A: For a standard Gaussian, the score points back toward zero.

$$p(x) = \mathcal{N}(0, 1) \implies s(x) = -x$$

Flashcard 7

Q: In a Gaussian distribution, what direction does a negative score at positive x point toward?

A: It points left, back toward the mean.

Flashcard 8

Q: Why does following the score field in a bimodal distribution steer samples toward modes?

A: Because the score points toward the nearest high-probability peak or mode.

Flashcard 9

Q: According to the denoising-score identity, what does the network noise prediction estimate?

A: It estimates the noise component and gives the score field at that noise level through the denoising-score identity.

$$s_t(x_t) \approx -\frac{\epsilon_{\theta}(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}$$

Flashcard 10

Q: What is the relationship between the score and predicted noise at timestep t ?

A: The score is approximately the negative predicted noise divided by the noising standard deviation.

$$s_t(x_t) \approx -\frac{\epsilon_{\theta}(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}$$

Flashcard 11

Q: What are the two primary components of the reverse diffusion sampling step?

A: A deterministic denoising drift and stochastic sampling randomness.

Flashcard 12

Q: In the reverse process, what is the purpose of adding calibrated noise at each step?

A: To sample from the full distribution rather than collapse to a single mode.

Flashcard 13

Q: Why is time embedding essential for the neural network in diffusion models?

A: The denoising behavior must change depending on whether the current sample is mostly noise or almost clean.

Flashcard 14

Q: In FiLM-style time conditioning, how does the time embedding modify internal feature maps?

A: It adds a learned time-dependent bias or modulation to each feature channel.

Flashcard 15

Q: What is the function of the reconstruction term in the diffusion ELBO?

A: It measures the model ability to reconstruct or predict clean data, such as x_0 from x_1 .

Flashcard 16

Q: What is the core idea of guided diffusion?

A: Modify the reverse drift or score field so samples move toward outputs consistent with a condition.

Flashcard 17

Q: How does classifier guidance mathematically modify the unconditional score?

A: It adds a classifier-gradient term to the unconditional score.

$$s^{\text{guided}}(x) = s(x) + w \nabla_x \log p_\phi(y \mid x, t)$$

Flashcard 18

Q: What does the parameter w represent in guided diffusion?

A: It is the guidance scale that balances conditioning strength against sample diversity.

Flashcard 19

Q: In CFG, how is unconditional behavior learned by the same model?

A: By randomly replacing the condition with a null token during training.

$$y \rightarrow \emptyset$$

Flashcard 20

Q: What is the formula for CFG noise prediction?

A: CFG combines conditional and unconditional outputs by extrapolating toward the condition.

$$\epsilon_{\text{CFG}} = (1 + w)\epsilon_{\text{cond}} - w\epsilon_{\text{uncond}}$$

Flashcard 21

Q: What is a primary advantage of CFG over classifier guidance?

A: CFG does not require an auxiliary classifier or classifier backpropagation during sampling.

Flashcard 22

Q: How does increasing guidance scale usually affect samples?

A: It improves condition alignment but reduces diversity and may introduce artifacts.

Flashcard 23**Q:** What does Fréchet Inception Distance measure?**A:** The distance between feature distributions of real and generated images.

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right)$$

Flashcard 24**Q:** Why is a noise-aware classifier required for standard classifier guidance?**A:** It must provide accurate gradients for noisy intermediate states at different timesteps.

$$p_\phi(y \mid x_t, t)$$

Flashcard 25**Q:** Concept: denoising drift.**A:** The deterministic part of the reverse step that pushes probability mass back toward higher-density regions.

$$\beta_t s_t(x_t)$$

Flashcard 26**Q:** Why can a very high guidance scale lead to artifacts?**A:** It can over-amplify condition-specific features and push samples away from the natural data manifold.**Flashcard 27****Q:** What is the null token in CFG?**A:** A placeholder condition used during training so the model learns an unconditional branch.

$$\emptyset$$

Flashcard 28**Q:** In the 1D bimodal example, what happens to the score between the two peaks?**A:** It becomes a weighted compromise based on how plausible each mode is at that position.

$$s(x) = \sum_k r_k(x) \left[-\frac{x - \mu_k}{\sigma^2} \right]$$

Flashcard 29**Q:** What is the Bayes foundation for classifier guidance?**A:** The conditional score equals the unconditional score plus the classifier score.

$$\nabla_x \log p(x \mid y) = \nabla_x \log p(x) + \nabla_x \log p(y \mid x)$$

Flashcard 30**Q:** How does evaluation mode affect diffusion sampling in PyTorch?**A:** It freezes Dropout and BatchNorm behavior, giving more stable score predictions.**Flashcard 31****Q:** Why is the score function considered a pull toward the expected value?**A:** It points in the direction of increasing probability, which leads toward the mean in a unimodal Gaussian.

Flashcard 32

Q: What is the relationship between noise prediction and variational inference in diffusion models?
A: Minimizing noise-prediction error is equivalent, up to constants and weights, to maximizing the diffusion ELBO.

$$\mathcal{L}_{\text{simple}} \leftrightarrow \mathcal{L}_{\text{ELBO}}$$

Flashcard 33

Q: The standard sampling loop starts with x_T sampled from which distribution?
A: Pure standard Gaussian noise.

$$x_T \sim \mathcal{N}(0, I)$$

Flashcard 34

Q: How does a deterministic reverse process with no added noise behave?
A: It behaves like gradient ascent on log-density and tends to collapse samples to modes.

Flashcard 35

Q: Why is it sensible to differentiate a classifier with respect to the reverse mean in guided diffusion?
A: The reverse mean is the model current proposal for the next image state, so its gradient tells how to adjust that proposal toward the condition.

Flashcard 36

Q: What does a lower FID value signify?
A: The generated feature distribution is closer to the real feature distribution, usually suggesting better realism and diversity match.

Flashcard 37

Q: In CFG training, if $p_{\text{drop}} = 0.1$, what percentage of the time does the model see the real label?
A: 90 percent.

$$1 - p_{\text{drop}} = 0.9$$

Flashcard 38

Q: What does the drift + diffusion update represent?
A: A discrete Langevin-style step combining score-guided drift with stochastic noise.

$$x_{t-1} = x_t + \beta_t s_t(x_t) + \sqrt{\beta_t} z$$

Flashcard 39

Q: In FiLM conditioning, why is the time embedding projected into different output sizes?
A: To match the channel widths of different convolutional blocks in the U-Net.

Flashcard 40

Q: What is the role of $\bar{\alpha}_t$ in the forward diffusion marginal?
A: It controls how much original signal is preserved at timestep t .

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

Flashcard 41

Q: How does the Gaussian score change as x moves farther from the mean?

A: The magnitude of the score increases linearly with distance from the mean.

$$s(x) = -\frac{x - \mu}{\sigma^2}$$

Flashcard 42

Q: What is the main tradeoff when using a small number of diffusion steps?

A: Sampling is faster, but image fidelity and guidance quality usually become weaker.

Flashcard 43

Q: In the noise-form reverse update, what term represents the predicted error that is subtracted?

A: The schedule-scaled predicted noise term.

$$\frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t)$$

Flashcard 44

Q: Why is normalization to a range such as $[-1, 1]$ important for diffusion stability?

A: It keeps the scale of data, predicted noise, and added Gaussian noise consistent.

Flashcard 45

Q: What happens if the classifier and diffusion model use inconsistent normalization?

A: The guidance gradient is in the wrong scale regime, making conditioning unreliable.

Flashcard 46

Q: In the ELBO decomposition, what do the intermediate KL terms ensure?

A: They ensure the learned reverse transition matches the true forward posterior.

$$p_\theta(x_{t-1} \mid x_t) \approx q(x_{t-1} \mid x_t, x_0)$$

Flashcard 47

Q: How does CFG extrapolate beyond the conditional model?

A: It pushes the prediction farther away from the unconditional prediction and toward the conditional one.

Flashcard 48

Q: What is denoising score matching basically doing?

A: Training a model to push noisy samples back toward high-density regions of the original data distribution.

Flashcard 49

Q: In the bimodal score derivation, what are responsibilities?

A: Weights showing how much each mode or component contributes to the total score at point x .

$$r_k(x) = \frac{p_k(x)}{\sum_j p_j(x)}$$

Flashcard 50

Q: Why do diffusion models start with x_T as pure noise?

A: The forward process is designed to destroy all data information until the final distribution is approximately standard Gaussian.

$$x_T \sim \mathcal{N}(0, I)$$

14. Last-Minute Exam Checklist

- Can you explain why $\mathcal{L}_{\text{simple}}$ is connected to the ELBO?
- Can you state the denoising-score identity with the correct square-root denominator?
- Can you explain why reverse diffusion needs both drift and noise?
- Can you compare classifier guidance and CFG in one paragraph?
- Can you explain why guidance scale w is a tradeoff, not a free improvement?
- Can you state what FID measures and why it is limited?

$$s_t(x_t) \approx -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}.$$

End of Chapter 4 study guide.